Learning with AI

Opportunities, limits, and open questions from AI-assisted research workflows

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Two genuine strengths

Information retrieval

- Like a far more powerful search engine
- Finds, summarizes, and connects sources fast
- Surfaces work you would never have found

Coding

- Like a strong, tireless programmer
- Writes, refactors, and debugs quickly
- Deploys an analysis in minutes, not days

On these two, AI is genuinely strong. Start there.

But a strength is not the whole job

Retrieval ≠ thinking

- Finding sources is not logical inference
- It is not creating a new argument
- Retrieval gives you inputs — not conclusions

Coding ≠ design

- Writing code is not empirical design
- The hard part is what to estimate, and why
- Code executes a design; it doesn't choose one

The strengths sit one level below the real work.

Two hurdles you cannot outsource

Make sense of the theory

- Different fields, different vocabularies for one problem
- Can you hold them together in your own mind?
- Integration is judgment — not retrieval

Comprehend the code

- Can you read the stack the agent built?
- Can you evaluate what it actually does?
- Can you defend every modeling choice in it?

AI raises the ceiling — only if you can reach it.

Organizing interdisciplinary literature

Many research topics are balkanized — one question, scattered across fields.

Sociology | **Political Science** | **Law & Society** | **Area Studies**

↓

One integrated picture — what AI's retrieval can help you assemble

AI can cross silos faster than any of us — if you can make sense of them.

Earl & Braithwaite (2022): repression research is "unproductively siloed" across disciplines.

Handle the empirical challenge properly

Even before AI: coding 2SLS (two-stage least squares) in R was easy. Deciding how to handle the endogeneity was the hard part.

Now, with AI: it can pick the model and write the code. Do you know what your results mean?

You can outsource coding, writing, even part of thinking — but not the deeper work of understanding.

“It can — but suboptimally”

Can AI agents do the harder tasks — inference, design — themselves?

~93%

of the way there — close to the conventional standard, but not at it. And maybe never.

(my argument — a working estimate, not a measured benchmark)

Close enough to rely on. Not close enough to stop thinking.

Cf. Goldfeder et al. (2026): the gap may be structural — specialized, adaptable systems (SAI), not general intelligence (AGI). arXiv:2602.23643.

Two moves that pay off

Retrieval → reach

- Expand your theoretical landscape
- Pull scholarship across fields together
- See the whole conversation, not your silo

Coding → speed

- Quickly deploy the design you have
- Evaluate the potential issues
- Report the findings back, fast

Use AI to widen what you reach and quicken what you test.

Practice 1 — ask the right questions

For human researchers

- Understand the theoretical context
- Understand the empirical design
- The question is the research

For AI

- ...the same understanding makes a better prompt
- Good question in → good work out
- Vague ask in → confident nonsense out

A good prompt is just a well-posed research question.

Practice 2 — read more, not less

Since I started working with AI (2021), I have read **more** theory — not less.

Why the counterintuitive direction?

- To ask better questions, I need deeper context
- To judge what AI surfaces, I need my own grounding
- To integrate across fields, I have to actually know them

AI handles the mechanical — which frees time for the conceptual.

Practice 3 — learn the unknowns AI surfaces

When the agent suggests something I don't know, that is the assignment — go learn it.

New data

a source you'd never have reached

New model

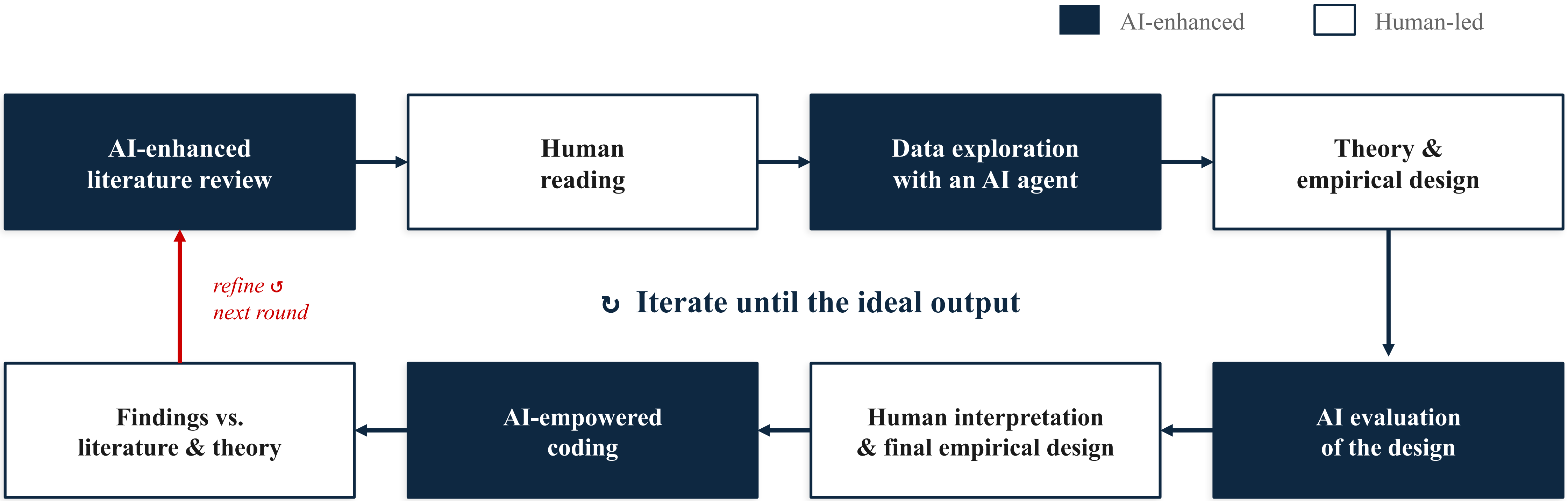
a specification you didn't know fit

New test

a robustness check worth running

Treat every unfamiliar suggestion as something to learn — not just to accept.

An AI-enhanced research loop



Human-led steps — reading, design, interpretation, the final draft — anchor every loop; AI accelerates the rest.

AI moves through every step — but you own the reading, the design, and the meaning.

The ideal output package (Liu et al. 2026)

I agree with their direction: redesign the artifact so nothing checkable is lost.

Two hidden taxes today

- Storytelling tax — failed experiments vanish to fit a linear narrative
- Engineering tax — prose drops the technical detail agents need

The fix — agent-native research artifacts:

- Scientific logic
- Fully-specified executable code
- **Exploration graph — keeps the dead ends & failures**
- Raw evidence behind every claim

Built for checking — agents answer questions about the paper far better (72.4% → 93.7%).

Keep the failures and the full detail — a paper built to be checked, by humans and agents.

Liu et al. (2026). "The Last Human-Written Paper: Agent-Native Research Artifacts." arXiv:2604.24658.

My robustness battery (CopChopper)

What "keep every check" looks like in one of my own projects:

```
copchopperGIS/results/
│
├── tables/comprehensive/   — robustness batteries
│      AppA_robust_strict            strict-window refit
│      AppB_robust_loiter            loiter outcome
│      AppC_robust_cell_eventmonth   cell × event-month FE
│      AppD_robust_h4_alternatives   alternative H4 specs
│      AppE_robust_misc              misc. sensitivity
│      AppF_h1_permutation           permutation inference
│
├── tables/individual/      — appendix battery (A1–A29):
│      Conley spatial SE · spatial spillover · PPML per agency ·
│      grid PPML · alt-grid · day-of-week & holiday FE · SMD balance ·
│      fleet roster & capacity · subsample · admin asymmetry · …
│
└── figures/                — event-study leads/lags · placebo-battery
       null distributions · family/admin forest plots · spec panels
```

Liu's ideal package, in miniature: keep every check — and ship it.

Source: copchopperGIS/results/ — every check is scripted, logged, and reproducible.

What means empirically rigid?

Reproducibility

- “The same-same test”
- Same data, same code
- Same result?

Robustness

- “The what-if test”
- Same data, other methods
- Conclusion survives?

Replicability

- “The new-data test”
- New data, same question
- Finding holds?

Most published work holds up — but checking it is slow human work AI can now carry.

These tests are mechanical by design — exactly where AI helps.

Brodeur et al. (2026), Nature 652:151–156: >85% of claims reproducible; 72% of estimates robust — across 110 articles.

What good AI-era research looks like

Two pillars — the understanding, and the proof:

Solid interdisciplinary vision & framing

- Cross-field theory, integrated — not siloed
- The understanding AI cannot supply
- This is what you bring to the table

A valid supporting package

- Reproducible, replicable, robust
- Built so others — and agents — can check
- The agent-native artifact we just saw

Framing carries the understanding; the package carries the proof.

Open questions — for this community

- ? Where does the durable ~93% gap actually sit — and does it move?
- ? How do we train comprehension when output is cheap?
- ? What should count as a valid supporting package now?
- ? Where does authorship — and responsibility — sit in an AI-assisted paper?

Thank you

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References

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Slides built with — and audited by — the multi-agent workflow described in this talk.

Opportunities · Limits · Questions

Opportunities

- Cross-silo theoretical reach
- Faster deploy–evaluate–report
- Verification carried at scale

Limits

- Retrieval ≠ inference
- Coding ≠ design
- A durable ~93% gap

Questions

- What is a valid package now?
- Where does authorship sit?
- How do we train comprehension?